
Synthetic-Like Content and Human Interaction in Online Communities

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Abstract

Synthetic-like text is becoming increasingly common across online communities, raising concerns about how it may affect authentic human interaction. While prior work has studied machine-generated text detection and online participation separately, less attention has been given to whether synthetic-like content prevalence is associated with changes in community-level interaction behavior. This study analyzes Reddit discussions from four communities over a twelve-week period to examine the relationship between estimated synthetic-like content and human-to-human reply activity. We build a pipeline for Reddit data collection, preprocessing, reply-chain reconstruction, manual annotation, text classification, and subreddit-week aggregation. We compare TF-IDF and sentence embedding features with Logistic Regression and XGBoost classifiers to estimate synthetic-likeness scores. Our results suggest that communities with higher estimated synthetic-like prevalence tend to show lower human-to-human reply interaction ratios. These findings highlight the importance of studying synthetic content not only as a detection problem, but also as a factor in online community health.

1 Introduction

The rapid adoption of generative artificial intelligence (AI) systems has significantly increased the volume of machine-generated and synthetic-like text appearing across online platforms. Large language models such as ChatGPT and similar systems are now widely used to generate posts, replies, summaries, and conversational content at scale [2, 1]. As synthetic text becomes increasingly integrated into digital communication environments, concerns have emerged regarding its potential influence on the quality, authenticity, and structure of online interaction.

Online communities rely heavily on authentic human-to-human engagement for information exchange, discussion, and social participation. Prior work has shown that interaction structures such as reply depth, branching behavior, and conversational continuity provide meaningful insight into the health and dynamics of online communities [5, 6]. At the same time, recent studies have investigated methods for detecting machine-generated text and have examined how generative AI may influence participation patterns in digital knowledge communities. However, existing work has largely studied these problems independently, and comparatively little attention has been given to whether higher levels of synthetic-like content are associated with measurable changes in community-level human interaction behavior.

In this work, we investigate the relationship between synthetic-like text prevalence and human interaction patterns within Reddit communities. Rather than attempting to determine definitive AI authorship, we focus on estimating synthetic-likeness as an observable textual characteristic and examine its association with reply-based interaction metrics. We analyze Reddit discussions

collected from four communities spanning expert-oriented and general-interest interaction styles over a twelve-week period.

To address these questions, we construct a multi-stage pipeline involving Reddit data collection, reply-chain reconstruction, manual annotation, text classification, and subreddit-week aggregation. We compare multiple classification pipelines using TF-IDF features, sentence embeddings, Logistic Regression, and XGBoost models to estimate synthetic-likeness scores for Reddit comments. Our findings suggest that communities with higher estimated synthetic-like prevalence tend to exhibit lower human-to-human reply interaction ratios across multiple subreddit types.

This work contributes to ongoing discussions surrounding generative AI and online community health by connecting synthetic-like textual behavior with broader interaction dynamics beyond standalone detection tasks.

2 Related Work

Research related to this work spans three primary areas: machine-generated text detection, online community interaction analysis, and the influence of generative AI systems on digital participation.

Prior work on machine-generated text detection has explored statistical, linguistic, and machine learning approaches for distinguishing synthetic text from human-written content. Early approaches focused on stylometric features and lexical patterns, while more recent work has increasingly adopted transformer-based methods and embedding-driven classification pipelines. Studies such as GLTR demonstrated that language-model token probability distributions can expose statistical irregularities in generated text [3], while later research explored supervised classifiers using contextual embeddings and transformer fine-tuning for AI-text detection tasks [7]. However, much of this literature focuses primarily on improving classification accuracy rather than examining downstream social effects.

A separate line of research has investigated interaction structures and participation dynamics within online communities. Prior studies have shown that reply depth, conversational branching, user engagement, and participation persistence provide meaningful indicators of community health and interaction quality [5, 6]. These approaches have been widely used in computational social science and social media mining to study information exchange, community formation, and discourse behavior.

More recently, researchers have begun examining how generative AI systems may influence online participation and communication practices. Existing studies have explored the use of large language models in online knowledge-sharing environments and social platforms. Concerns have emerged regarding misinformation amplification, low-effort content generation, and the increasing prevalence of synthetic or semi-synthetic communication online [1, 4].

Our work differs from prior research by combining machine learning-based synthetic-likeness estimation with subreddit-level temporal aggregation and reply-based interaction analysis to study broader patterns of community behavior.

3 Data and Methodology

3.1 Data Collection

Reddit was selected as the primary platform for analysis due to its threaded discussion structure, publicly accessible conversational data, and diversity of community interaction styles. Data was collected from four subreddits representing both expert-oriented and general-interest communities: *AskScience*, *AskDocs*, *Advice*, and *TodayILearned*. These communities were selected to capture variation in conversational norms, moderation styles, and interaction behavior.

The dataset was collected over a fixed twelve-week period spanning October 6, 2025 through December 28, 2025. A custom Python-based collection pipeline was developed to retrieve Reddit posts and comments through structured API requests while handling pagination and rate-limiting constraints. For each post and comment, metadata including identifiers, timestamps, parent-child relationships, and textual content were stored for downstream analysis.

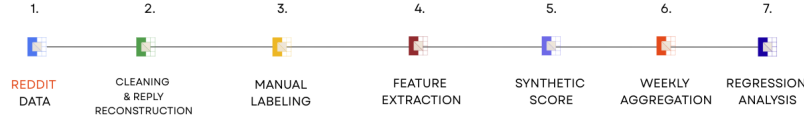


Figure 1: Overview of the synthetic-likeness analysis pipeline, including Reddit data collection, preprocessing, manual annotation, feature extraction, synthetic scoring, aggregation, and regression analysis.

To maintain temporal consistency across communities, approximately balanced weekly collection targets were used for each subreddit. The final cleaned corpus contained 7,606 comments aggregated into 48 subreddit-week observations.

3.2 Data Cleaning and Reply Reconstruction

Several preprocessing and filtering steps were applied prior to analysis. Deleted, removed, and blank comments were excluded from the dataset. AutoModerator-generated responses, repeated templates, moderator boilerplate, and meta-discussion threads were also removed to reduce distortion from non-organic conversational behavior.

Parent-child reply relationships were reconstructed to distinguish direct post replies from comment-to-comment interactions. Broken reply chains with missing parent context were pruned when reliable reconstruction was not possible. Following preprocessing, cleaned weekly summary files and annotation packets were generated for downstream labeling and classification tasks.

3.3 Synthetic-Likeness Modeling

Rather than attempting definitive AI authorship detection, this study focuses on estimating *synthetic-likeness* as a textual characteristic. Comments were manually annotated using three labels: human-like, synthetic-like, and unsure. The resulting annotations were used to construct supervised classification pipelines for downstream synthetic-likeness estimation.

Two feature representations were evaluated: TF-IDF features and sentence embeddings generated using the all-MiniLM-L6-v2 embedding model. Four classification pipelines were compared:

- TF-IDF + Logistic Regression
- TF-IDF + XGBoost
- Sentence Embeddings + Logistic Regression
- Sentence Embeddings + XGBoost

To reduce leakage between related comments, train-test splits were grouped by post identifier rather than randomized at the individual comment level. Models were evaluated using accuracy, F1 score, and ROC-AUC metrics.

3.4 Interaction Metrics and Aggregation

Two primary weekly metrics were computed for each subreddit-week observation.

The first metric, *synthetic prevalence*, was defined as the mean synthetic-likeness score across comments within a subreddit-week aggregation.

The second metric, *human-to-human interaction ratio (h2h ratio)*, measured the proportion of reply interactions where both the parent comment and reply comment were classified as human-like. A threshold of 0.5 was used as the operational split between human-like and synthetic-like classifications.

Weekly subreddit-level metrics were aggregated across the twelve-week collection period to enable comparative community-level analysis.

Table 1: Performance comparison of synthetic-likeness classification pipelines.

Pipeline	Accuracy	F1	ROC-AUC
Embeddings + Logistic Regression	0.817	0.329	0.849
Embeddings + XGBoost	0.937	0.261	0.825
TF-IDF + Logistic Regression	0.925	0.167	0.776
TF-IDF + XGBoost	0.902	0.087	0.673

3.5 Regression Analysis

To examine the relationship between synthetic-like prevalence and human interaction, a regression model was constructed using subreddit-week observations as the unit of analysis. The primary model specification was:

$$h2h_ratio \sim synthetic_prevalence + C(subreddit)$$

where subreddit categorical controls were included to account for baseline differences across community types. The project repository, preprocessing scripts, analysis notebooks, and supporting materials are publicly available at:

<https://github.com/zinels/synthetic-content-human-interaction>

4 Results

4.1 Classification Performance

Four classification pipelines were evaluated for synthetic-likeness estimation using combinations of TF-IDF features, sentence embeddings, Logistic Regression, and XGBoost models. Performance was evaluated on held-out data using accuracy, F1 score, and ROC-AUC metrics.

As shown in Table 1, the strongest overall performance was achieved by the Sentence Embeddings + Logistic Regression pipeline, which obtained an accuracy of 0.817, F1 score of 0.329, and ROC-AUC of 0.849. Embedding-based models generally achieved stronger F1 and ROC-AUC performance than TF-IDF baselines, supporting the hypothesis that contextual semantic representations are more effective for estimating synthetic-like textual behavior.

4.2 Synthetic Prevalence and Human Interaction

Regression analysis identified a strong negative relationship between estimated synthetic-like prevalence and human-to-human interaction ratio across subreddit-week observations. The coefficient associated with synthetic prevalence was negative ($\beta = -1.491$) and statistically significant ($p < 0.001$), with an overall model fit of $R^2 = 0.863$.

As shown in Figure 2, visual analysis of subreddit-week observations revealed a consistent downward trend between synthetic prevalence and reply-based human interaction. Communities with lower estimated synthetic prevalence tended to exhibit higher levels of human-to-human reply engagement, while communities with higher estimated synthetic prevalence generally showed lower interaction ratios.

Distinct clustering behavior was also observed across subreddit communities. Expert-oriented communities and general-interest communities exhibited different interaction profiles, although the overall negative relationship between synthetic prevalence and human interaction remained visible across subreddit types.

4.3 Subreddit-Level Patterns

As shown in Figure 3, subreddit-level aggregation revealed consistent ordering patterns across communities. *TodayILearned* exhibited the lowest average synthetic prevalence and the highest

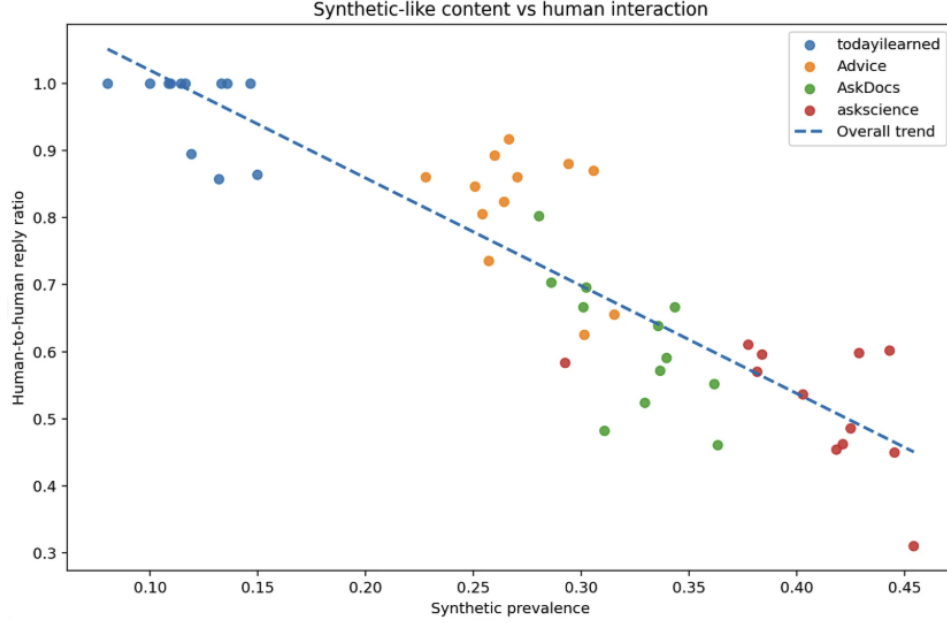


Figure 2: Relationship between synthetic-like prevalence and human-to-human reply interaction ratio across subreddit-week observations. The dashed regression line illustrates the overall negative association between estimated synthetic prevalence and reply-based human interaction.

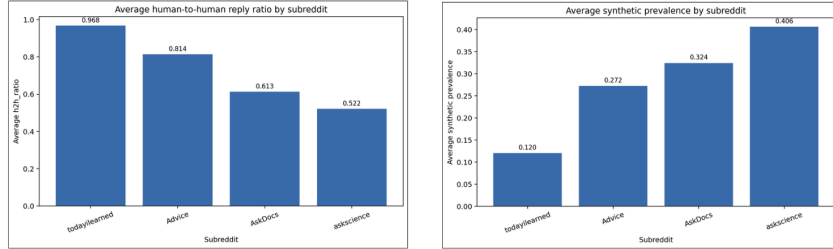


Figure 3: Average human-to-human interaction ratios and estimated synthetic prevalence across subreddit communities.

human-to-human interaction ratio, while *AskScience* showed the highest estimated synthetic prevalence and the lowest human-to-human interaction ratio.

The intermediate communities, *Advice* and *AskDocs*, occupied middle positions along both dimensions, suggesting a gradient-like relationship between synthetic prevalence and reply-based engagement behavior.

Across all four subreddit communities, higher estimated synthetic prevalence was consistently associated with lower levels of human-to-human reply interaction. These findings support the primary hypothesis that communities with greater synthetic-like textual behavior may exhibit weaker reply-based conversational engagement.

5 Discussion

The findings of this study suggest that higher estimated synthetic-like content prevalence is associated with lower levels of human-to-human reply interaction across multiple Reddit communities. While this work does not establish causality or definitive AI authorship detection, the observed relationship indicates that synthetic-like textual behavior may correspond with measurable shifts in conversational engagement patterns within online communities.

One interpretation is that communities with higher concentrations of synthetic-like or low-effort generated content may experience weaker conversational engagement and reduced reply depth, as users become less likely to participate in extended discussion.

The observed subreddit-level ordering patterns further support this interpretation. Communities such as *TodayILearned*, which exhibited lower estimated synthetic prevalence, also demonstrated higher human-to-human interaction ratios. In contrast, communities with higher estimated synthetic prevalence showed comparatively weaker reply-based engagement behavior.

This work also contributes to existing literature by connecting machine-generated text analysis with broader computational social science perspectives on online interaction. Prior research has often studied generated text detection, online participation trends, and conversational structure separately [3, 5, 8]. In contrast, this study combines synthetic-likeness estimation with reply-network interaction analysis at the subreddit-week level, allowing synthetic textual behavior to be examined alongside community interaction dynamics rather than solely as an isolated classification problem.

Several limitations should be acknowledged. The classifier estimates stylistic similarity rather than definitive AI authorship, and overall F1 performance remained modest. Additionally, the dataset was limited to four Reddit communities collected over a twelve-week period, limiting broader generalization.

Future work could expand this analysis through larger multi-platform datasets, stronger transformer-based classifiers, longitudinal analysis, and causal inference approaches.

Despite these limitations, the findings provide preliminary evidence that synthetic-like textual behavior may be associated with measurable changes in reply-based human interaction within online communities. More broadly, this work highlights the importance of studying generative AI not only as a text-generation technology, but also as a potential factor influencing the structure and dynamics of online social spaces.

6 Conclusion

This study investigated the relationship between synthetic-like textual behavior and human interaction dynamics within Reddit communities. Using a multi-stage pipeline involving Reddit data collection, reply-chain reconstruction, manual annotation, text classification, and subreddit-week aggregation, we examined whether estimated synthetic-like content prevalence was associated with changes in reply-based human interaction.

Across multiple subreddit communities, higher estimated synthetic-like prevalence was consistently associated with lower human-to-human interaction ratios. While the analysis does not establish causality or definitive AI authorship, the findings suggest that synthetic-like textual behavior may correspond with measurable differences in online conversational engagement patterns.

Beyond text classification alone, this work contributes to ongoing discussions surrounding generative AI and online community health by connecting synthetic-likeness estimation with broader interaction dynamics. As generative AI systems continue to become integrated into online communication environments, understanding their potential influence on participation, discourse structure, and community behavior will remain an increasingly important area of research in computational social science and social media mining.

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